

Only if Battery already includes Battery Protection: Cut JP1, Bridge JP2, DNP: R1:4, Q1:2, C1, U1

J3
 Molex Picoblade 6P

VBus
 CC1
 CC2
 D-
 D+
 GND

6
 5
 4
 3
 2
 1

R22 22R
 R23 22R

D-
 D+

TP20

C2 100nF

R5 5.1k
 R6 5.1k

U3
 USBLC6-4SC6

GND

CC1 and CC2 pulled low = USB-Device
 Keep tracks to U3 short, place C2 close to U3
 Place R22:23 close to U10

VPCC function disabled

R7 0R 10µF

GND

U2 MCP73871-2CC

R11 10k R12 10k R13 10k

R8 DNP

JP4

JP3

C5 4.7µF

C4 4.7µF

GND

PROG1 13

PROG2 4

PROG3 12

THERM 5

TE 9

CE 17

Bat_NTC

VCC

Vout

Vout

BMS_Pg

BMS_Sht1

BMS_Sht2

VPowerpath

CC-Charging up to 500mA (5 USB Unit Loads)

CV-Charging until charge current drops to 50mA

JP3;4 are for Testing purposes, use them to isolate the battery management from the rest of the circuit

Interface not used by default

Includes: I2C Pins, UART (or 2 ESP32 GPIOs), 3 free GPIOs, 1 free TCAB418 GPIO (only slower signals recommended), ESP_ON for powerup

Always check I2C Address compatibility when connecting new I2C devices

[illegible]

Note: Data Pins are denoted so they match with the Espressif Camera Component, where "D0" is D2. "D1" equals D3 etc.

[illegible][illegible][illegible]

The diagram illustrates the hardware connection for the Keypad module. The Keypad module is connected to the TC595 IC (U9) via I2C (SCL, SDA) and a Reset line. The Keypad module has a Keypad_Enable pin connected to the Reset pin of the TC595. The Keypad module also has a Keypad_Int pin connected to the INT pin of the TC595. The Keypad module is powered by +3V3 and has a GND connection. The TC595 IC is labeled U9 and has pins for VCC, GND, and I2C addresses (ROW0-ROW7, COL0-COL7). The I2C addresses are connected to BMS_Stat1, BMS_PC, and FreeGPIO_0+.

Rev: 0
Id: 1/1